CypherVault - Secure Image Encryption

Project Overview

This project is a Flask-based secure image encryption system that demonstrates hybrid crypto graphy using AES-256-GCM for image encryption and RSA-OAEP for wrapping the AES key. It is d esigned for secure storage and transfer of sensitive image content through a web application.

Key Features

- AES-256-GCM encrypted image handling
- RSA key wrapping and secure transfer model
- Flask web app with dashboard and UI flow
- SQLAlchemy database models and secure password hashing
- Security headers and validation for file handling
- Project testing with pytest (19 tests passing)

Technology Stack

- Python 3.14
- Flask, Flask-SQLAlchemy, Flask-Login, Flask-WTF, Flask-Limiter
- cryptography, PyCryptodome, Pillow
- SQLite database and HTML/CSS/JS frontend

Validation

The project was verified locally with pytest and responded successfully at http://127.0.0.1:5000 with HTTP 200. The health endpoint returned status ok and database ok.

Submission Summary

This project demonstrates secure image encryption, secure key management,
and a web-based interface suitable for academic submission and presentation.